

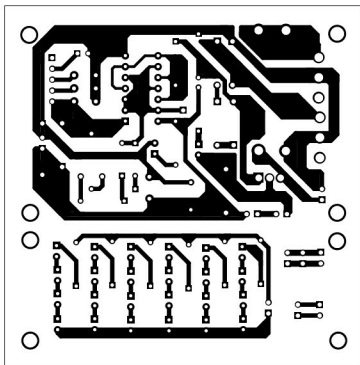


Fig. 2: An actual-size, single-side PCB for the simple manual PWM generator

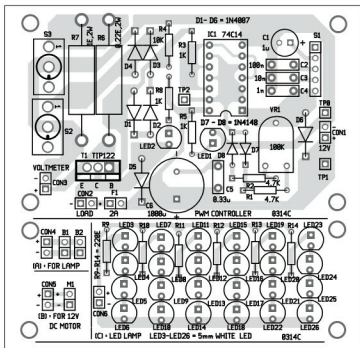


Fig. 3: Component layout for the PCB

T1. If the voltage drop over these diodes is above $3 \times 0.7V = 2.1V$ with S2 closed or $4 \times 0.7V = 2.8V$ with S2 open, these diodes limit the base current of T1 and protect it from destruction. If you omit D1-D4 and a short circuit occurs at CON2, fuse F1 will blow and transistor T1 probably will get destroyed. Resistors R6 and R7 and switch S3 determine the maximum current (the threshold) of the current limiting function accomplished by D1-D4.

on 12V and current rating is around 300mA

3. Load with 24 white bright LEDs

To test the circuit for proper functioning, follow test points table and verify all voltage levels accordingly. ●

The author is a regular contributor to EFX. He was a researcher and assistant professor in Technical University of Sofia (Bulgaria) and an expert lecturer in Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria

Construction and testing

An actual-size, single-side PCB for simple manual PWM generator is shown in Fig. 2 and its component layout in Fig. 3.

After assembling the circuit on PCB, enclose it in a suitable plastic case. Fix all the connectors and switches on PCB and make them accessible from outside the case. CON3 may be used to connect a multimeter to measure the voltage across resistors R6 and R7, if required.

The loads can be mounted on the PCB itself. To connect any load, just connect CON2 to the corresponding load connector such as CON5 for DC motor. The loads are:

1. Load with two lamps that works on 12V. Power is 5W and current is around 400mA

2. Load with DC motor works

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